

Astro vulnerable to URL manipulation via headers, leading to middleware and CVE-2025-61925 bypass

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- Published: 2025-11-13
- Original: <<https://github.com/withastro/astro/security/advisories/GHSA-hr2q-hp5q-x767>>
- Preserved from: <https://github.com/withastro/astro/security/advisories/GHSA-hr2q-hp5q-x767> (github-api) on 2026-10-02
- Licence: unknown

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Content

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Astro vulnerable to URL manipulation via headers, leading to middleware and CVE-2025-61925 bypass

- Advisory: GHSA-hr2q-hp5q-x767
- CVE: CVE-2025-64525
- Severity: medium
- Published: 2025-11-13
- Updated: 2025-11-13

Affected

- `astro` (npm): $\geq 2.16.0$, $< 5.15.5$, fixed in 5.15.5

Description

Summary

In impacted versions of Astro using [on-demand rendering](#), request headers `x-forwarded-proto` and `x-forwarded-port` are insecurely used, without sanitization, to build the URL. This has several consequences the most important of which are:

- Middleware-based protected route bypass (only via `x-forwarded-proto`)
- DoS via cache poisoning (if a CDN is present)
- SSRF (only via `x-forwarded-proto`)
- URL pollution (potential SXSS, if a CDN is present)
- WAF bypass

Details

The `x-forwarded-proto` and `x-forwarded-port` headers are used without sanitization in two parts of the Astro server code. The most important is in the `createRequest()` function. Any configuration, including the default one, is affected:

<https://github.com/withastro/astro/blob/970ac0f51172e1e6bff4440516a851e725ac3097/packages/astro/src/core/app/node.ts#L97> <https://github.com/withastro/astro/blob/970ac0f51172e1e6bff4440516a851e725ac3097/packages/astro/src/core/app/node.ts#L121>

These header values are then used directly to construct URLs.

By injecting a payload at the protocol level during URL creation (via the `x-forwarded-proto` header), the entire URL can be rewritten, including the host, port and path, and then pass the rest of the URL, the real hostname and path, as a query so that it doesn't affect (re)routing.

If the following header value is injected when requesting the path `/ssr`:

```
x-forwarded-proto: https://www.malicious-url.com/?tank=
```

The complete URL that will be created is: `https://www.malicious-url.com/?tank=//localhost/ssr`

As a reminder, URLs are created like this:

```
url = new URL(`${protocol}://${hostnamePort}${req.url}`);
```

The value is injected at the beginning of the string (`${protocol}`), and ends with a query `?tank=` whose value is the rest of the string, `://${hostnamePort}${req.url}`.

This way there is control over the routing without affecting the path, and the URL can be manipulated arbitrarily. This behavior can be exploited in various ways, as will be seen in the PoC section.

The same logic applies to `x-forwarded-port`, with a few differences.

[!NOTE] The `createRequest` function is called every time a non-static page is

requested. Therefore, all non-static pages are exploitable for reproducing the attack.

PoC

The PoC will be tested with a minimal repository:

- Latest Astro version at the time (2.16.0)
- The Node adapter
- Two simple pages, one SSR (/ssr), the other simulating an admin page (/admin) protected by a middleware
- A middleware example copied and pasted from the official Astro documentation to protect the admin page based on the path

[Download the PoC repository](#)

Middleware-based protected route bypass - x-forwarded-proto only

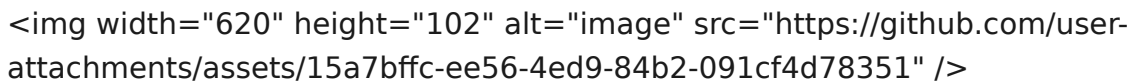
The middleware has been configured to protect the /admin route based on [the official documentation](#):

```
// src/middleware.ts
import { defineMiddleware } from "astro/middleware";

export const onRequest = defineMiddleware(async (context, next) => {
  const isAuthenticated = false; // auth logic
  if (context.url.pathname === "/admin" && !isAuthenticated) {
    return context.redirect("/");
  }
  return next();
});
```

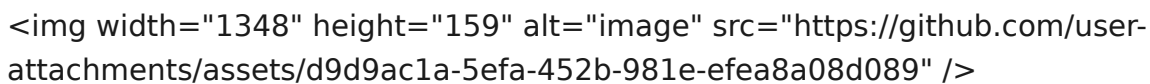
1. When trying to access /admin the attacker is naturally redirected :

```
curl -i http://localhost:4321/admin
```



1. The attacker can bypass the middleware path check using a malicious header value:

```
curl -i -H "x-forwarded-proto: x:admin?" http://localhost:4321/admin
```



How is this possible?

Here, with the payload `x:admin?`, the attacker can use the URL API parser to their advantage:

- `x:` is considered the protocol
- Since there is no `//`, the parser considers there to be no authority, and everything before the `?` character is therefore considered part of the path: `admin`

During a path-based middleware check, the *path* value begins with a `/`:

`context.url.pathname === "/admin"`. However, this is not the case with this payload;

`context.url.pathname === "admin"`, the absence of a slash satisfies both the middleware check and the router and consequently allows us to bypass the protection and access the page.

SSRF

As seen, the request URL is built from untrusted input via the `x-forwarded-protocol` header, if it turns out that this URL is subsequently used to perform external network calls, for an API for example, this allows an attacker to supply a malicious URL that the server will fetch, resulting in server-side request forgery (SSRF).

Example of code reusing the "origin" URL, concatenating it to the API endpoint :

```

```

DoS via cache poisoning

If a CDN is present, it is possible to force the caching of bad pages/resources, or 404 pages on the application routes, rendering the application unusable.

A `404` can be forced, causing an error on the `/ssr` page like this : `curl -i -H "x-forwarded-proto: https://localhost/vulnerable?" http://localhost:4321/ssr `

Same logic applies to `x-forwarded-port` : `curl -i -H "x-forwarded-port: /vulnerable?" http://localhost:4321/ssr`

How is this possible?

The router sees the request for the path `/vulnerable`, which does not exist, and therefore returns a `404`, while the potential CDN sees `/ssr` and can then cache the `404` response, consequently serving it to all users requesting the path `/ssr`.

URL pollution

The exploitability of the following is also contingent on the presence of a CDN, and is therefore cache poisoning.

If the value of `request.url` is used to create links within the page, this can lead to Stored XSS with `x-forwarded-proto` and the following value:

```
x-forwarded-proto: javascript:alert(document.cookie)//
```

results in the following URL object:

```

```

It is also possible to inject any link, always, if the value of `request.url` is used on the server side to create links.

```
x-forwarded-proto: https://www.malicious-site.com/bad?
```

The attacker is more limited with `x-forwarded-port`

If the value of `request.url` is used to create links within the page, this can lead to broken links, with the header and the following value:

```
X-Forwarded-Port: /nope?
```

Example of an Astro website: ``

WAF bypass

For this section, Astro invites users to read previous research on the React-Router/Remix framework, in the section "Exploitation - WAF bypass and escalations". This research deals with a similar case, the difference being that the vulnerable header was `x-forwarded-host` in their case:

<https://zhero-web-sec.github.io/research-and-things/react-router-and-the-remixed-path>

Note: A section addressing DoS attacks via cache poisoning using the same vector was also included there.

CVE-2025-61925 complete bypass

It is possible to completely bypass the vulnerability patch related to the `X-Forwarded-Host` header.

By sending `x-forwarded-host` with an empty value, the `forwardedHostname` variable is assigned an empty string. Then, during [the subsequent check](#), the condition fails because `forwardedHostname` returns `false`, its value being an empty string:

```
if (forwardedHostname && !App.validateForwardedHost(...))
```

Consequently, the implemented check is bypassed. From this point on, since the request has no `host` (*its value being an empty string*), the path value is retrieved by the URL parser to set it as the `host`. This is because the `http/https` schemes are considered special schemes by the [WHATWG URL Standard Specification](#), requiring an `authority state`.

From there, the following request on the example SSR application (astro repo) yields an SSRF: `` empty `x-forwarded-host` + the target `host` in the path

Credits

- Allam Rachid ([zhero](#);))
- Allam Yasser (inzo)

References

- `<https://github.com/withastro/astro/security/advisories/GHSA-hr2q-hp5q-x767>`
- `<https://nvd.nist.gov/vuln/detail/CVE-2025-64525>`
- `<https://github.com/withastro/astro/commit/dafbb1ba29912099c4faff1440033edc768af8b4>`
- `<https://github.com/withastro/astro/blob/970ac0f51172e1e6bff4440516a851e725ac3097/packages/astro/src/core/app/node.ts#L121>`
- `<https://github.com/withastro/astro/blob/970ac0f51172e1e6bff4440516a851e725ac3097/packages/astro/src/core/app/node.ts#L97>`
- `<https://github.com/advisories/GHSA-hr2q-hp5q-x767>`

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